Input Reference

This section documents every SWAT+ input file. Each page lists:

- the file's purpose
- the Fortran reader and the derived type it populates
- the column layout, units, defaults
- a real example from the reference datasets
- known bugs or quirks in an `Important` section at the bottom

Every page is verified against the SWAT+ source at a specific commit, recorded in the page frontmatter as `verified_at_commit`. The source code is the only authoritative reference; example datasets are downstream and can drift.

Simulation control

- `file.cio`. Master index. Names every other input file.
- `time.sim`. Simulation period and time step.
- `print.prt`. Output selection and aggregation.
- `object.cnt`. Basin name, areas, spatial object counts.

Basin parameters

- `codes.bsn`. Method switches.
- `parameters.bsn`. Numeric coefficients.

Spatial objects and connectivity

- `*.con`. Connectivity files (hru.con, chandeg.con, aquifer.con, reservoir.con, …).
- `hru-data.hru`. Per-HRU pointer table.
- `*.hyd`, `field.fld`, `shade_factor.shf`. Per-HRU hydrology, topography, field geometry, and shade factor. Referenced by `hru-data.hru`.
- `recall`. Point sources and inlets. Master `recall.rec` plus salt and constituent side databases and the region/element files.

Structural BMPs

- `*.str`, `septic.sep`, `urban.urb`. Tile drains, septic systems and effluent, filter strips, grassed waterways, user-defined BMPs, saturated buffers, urban land cover.

Climate

- `*.cli`. Index files: weather-sta.cli, weather-wgn.cli, pcp.cli, tmp.cli, slr.cli, hmd.cli, wnd.cli, pet.cli, atmodep.cli.
- `*.pcp`. Precipitation data files.
- `*.tmp`. Temperature data files.
- `*.tem`. Legacy temperature file extension.
- `snow.sno`. Snow parameter database.

Soils and land use

- soils.sol. Soil profile and layer properties.
- landuse.lum. Land-use management.
- *.lum lookup tables. Curve number, conservation practice, and overland-flow Manning's n tables referenced by landuse.lum.

Initial conditions

- *.ini. General initial conditions: plant.ini (plant communities), soil_plant.ini (soil water, residue, plant biomass), om_water.ini (organic matter in water bodies). Solute-side init files are covered in their module pages under Solute transport and Pollutants.

Channels, aquifers, reservoirs, wetlands

- *.cha. Legacy channel files.
- sd_channel. SWAT+ routing channel inputs (hydrology.cha, sediment.cha, nutrients.cha).
- *.aqu. Aquifer files (aquifer.aqu, initial.aqu).
- *.res. Reservoir files.
- *.wet. Wetland files.

Groundwater

- gwflow.*. 2D cell-based groundwater module inputs (mesh, channel coupling, tiles, ponds, canals, solutes, heat).

Operation and management databases

- management.sch and minor inputs. Fixed-schedule management, puddle.ops, transplant.plt, co2.out, object.prt.
- *.ops. harv, graze, irr, chem_app, fire, sweep.
- *.dtl. Decision tables (lum.dtl, res_rel.dtl, scen_lu.dtl, flo_con.dtl).
- plants.plt. Plant parameter database.
- fertilizer.frt. Fertilizer composition and the manure type and organic-matter databases (manure_db.frt, manure_om.frt).
- manure_allo.mnu. Manure allocation. Sources, demands, and the rule that distributes manure between them.
- tillage.til. Tillage operations.
- carb_coefs.cbn. Century carbon coefficients.
- treatment.trt. Legacy treatment placeholder.

Water allocation

- water_allocation.wro and the *.wal transport files. Water-rights and inter-object water transfer module: master allocation file plus canal, pipe, tower, treatment, use, out-of-basin source and receiver inputs.

Solute transport

- salt module. rtb-salt inputs: per-object initial conditions and applied loads for the eight salt ions.
- cs module. rtb-cs inputs: master constituents.cs plus per-object initial conditions and applied loads for selenium, boron, and other constituents.

Pollutants

- pesticide module. pesticide.pes parameter database, per-HRU and per-channel initial conditions, and the parent-daughter metabolite chain.
- pathogen module. pathogens.pth parameter database and per-HRU and per-channel initial conditions.
- metals module. Per-HRU initial heavy metal concentrations.

Verification procedure

The procedure for verifying a new or updated page is in SOURCE-OF-TRUTH.md in the docs working folder.